

GabayPoz Recommender System — Version 1.2

Demonstration of Program-First Recommendations with Q7 Strand Multipliers

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Abstract

This document is an executable demonstration of the GabayPoz recommender, version 1.2. It is generated by running `demo.py` against the bundled sample fixtures under `data/sample/` and emitting a LaTeX fragment (`results.tex`) that this wrapper includes verbatim. Each subsection describes one synthetic student profile, including Q7 strand, Q10 affordability, Q11 mobility, Q12 duration tolerance, the top-three program recommendations, the primary school suggestion, and the rank-1 explanation.

1. About this demonstration

GabayPoz is a constraint-aware, explainable college program recommender for Grade 11/12 students in Pozorrubio, Pangasinan. EDA established that tertiary access is shaped strongly by geography and household economics, so the recommender keeps Q11 commute and Q10 affordability as hard school filters. Version 1.2 also restores Q7 as a strand-based aptitude multiplier, using the tested v1 multiplier map.

The current demo pipeline is:

1. **Student profile.** Q1–Q6 build internal affinity; Q8–Q9 build aptitude; Q7 multiplies matching aptitude dimensions only.
2. **Program ranking.** Programs are ranked first using student fit, municipality saturation as small context, and the Q12 penalty.
3. **School selection.** For each selected program, feasible schools are filtered by Q11 commute and Q10 affordability.
4. **Persistence shape.** The model emits exactly three ranked rows using the v1.2 model ID; explanations and alternate schools remain in the API response.

The four profiles below exercise STEM, HUMSS, TVL, and GAS strand inputs.

2. Demonstrative profiles

D1: Health-oriented STEM strand, ready for board-exam programs

Model: `tds_recommender_v1_2`; Barangay id: 1; Q7: STEM; Q10: C; Q11: C; Q12: A.

Top recommendations.	Rank	Program	Primary school	Score
	1	BS Nursing	PSU Lingayen	0.746
	2	BS Medical Technology	PSU Lingayen	0.669
	3	Bachelor of Elementary Education	PSU Lingayen	0.198

Rank-1 explanation. BS Nursing matches your strongest Health signals. PSU Lingayen is the primary school suggestion because it passed your travel and affordability limits. Local Health field presence is included only as small context, not a job guarantee.

D2: Arts-oriented HUMSS strand, nearby-only travel

Model: `tds_recommender_v1_2`; Barangay id: 1; Q7: HUMSS; Q10: B; Q11: A; Q12: B.

	Rank	Program	Primary school	Score
Top recommendations.	1	AB Communication	PSU Lingayen	0.760
	2	BS Business Administration	PSU Lingayen	0.328
	3	BS Computer Science	PSU Lingayen	0.195

Rank-1 explanation. AB Communication matches your strongest Arts signals. PSU Lingayen is the primary school suggestion because it passed your travel and affordability limits. Local Arts field presence is included only as small context, not a job guarantee.

D3: Agriculture-oriented TVL strand, moderate budget

Model: `tds_recommender_v1_2`; Barangay id: 3; Q7: TVL; Q10: B; Q11: B; Q12: B.

	Rank	Program	Primary school	Score
Top recommendations.	1	BS Agriculture	PSU Lingayen	0.825
	2	BS Civil Engineering	PSU Lingayen	0.168
	3	BS Medical Technology	PSU Lingayen	0.159

Rank-1 explanation. BS Agriculture matches your strongest Agriculture signals. PSU Lingayen is the primary school suggestion because it passed your travel and affordability limits. Local Agriculture field presence is included only as small context, not a job guarantee.

D4: STEM-oriented GAS strand, moderate budget

Model: `tds_recommender_v1_2`; Barangay id: 2; Q7: GAS; Q10: B; Q11: B; Q12: B.

	Rank	Program	Primary school	Score
Top recommendations.	1	BS Computer Science	PSU Lingayen	0.775
	2	BS Civil Engineering	PSU Lingayen	0.672
	3	BS Medical Technology	PSU Lingayen	0.393

Rank-1 explanation. BS Computer Science matches your strongest STEM signals. PSU Lingayen is the primary school suggestion because it passed your travel and affordability limits. Local STEM field presence is included only as small context, not a job guarantee.

3. How to reproduce this PDF

```
cd demo
make                # runs demo.py then pdflatex twice
```

`make clean` removes the LaTeX intermediates and the generated `results.tex`; `make distclean` additionally removes `demo.pdf`. Re-running from scratch reproduces the PDF deterministically: the

recommender is fully stateless, the seven sample CSVs are version-controlled, and tie-breaking is documented in TDS §5 Phase 5.

4. References

- **TDS:** docs/tds.md — the authoritative specification this implementation traces to.
- **EDA v1:** docs/eda_v1.pdf — Pangasinan education, employment, and access profile.
- **EDA v2:** docs/eda_v2.pdf — Pozorrrubio program access, scholarship analysis, and employment signals.
- Burke, R. (2002). Hybrid recommender systems. *User Modeling and User-Adapted Interaction*, 12(4).
- Liu, T.-Y. (2009). Learning to rank for information retrieval. *Foundations and Trends in IR*, 3(3).
- Philippine Institute for Development Studies (2019). *Tracer study of CHED-scholarship and non-scholarship graduates* (Discussion Paper No. 2019-26).